



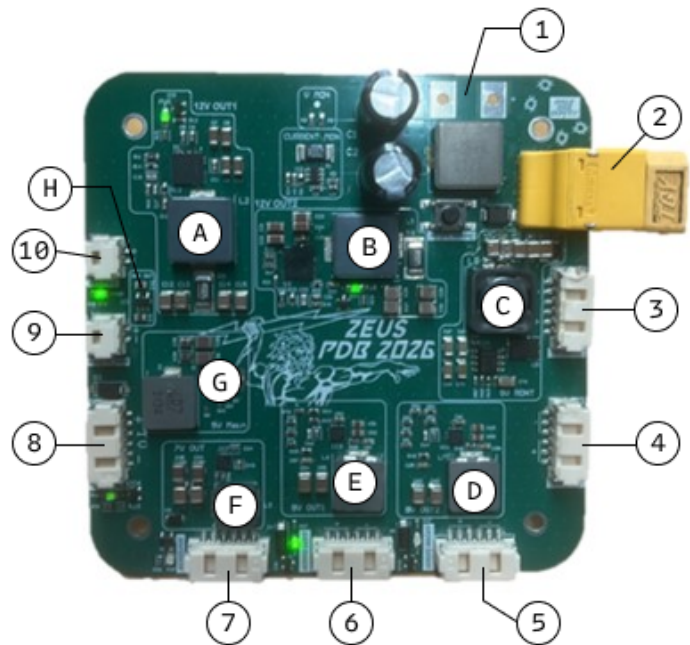
"ZEUS" PDB V1

1 Features

- Wide input voltage range: 16.8 V to 50.4 V
- Safety factor of 1.25
- Harmonic filtering with overvoltage protection
- Current and voltage monitor
- 12S and 4S input support
- No dedicated avionics battery required
- Onboard telemetry output

2 Applications

- Fixed wing UAV
- Flight controller supply
- Control-surface servo power
- Payload, video and datalink
- Vision camera
- Current and voltage telemetry

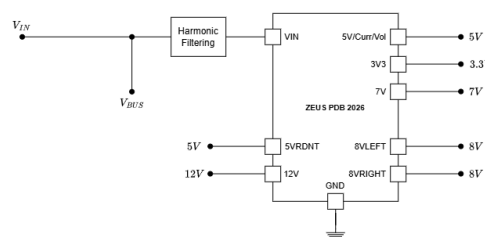


3 Description

The power distribution board (PDB) is a multi-board device that takes the raw power from LiPo batteries and distributes it to the various peripherals on the aircraft. The goal of Avionics PDB 2026 is to create a versatile PDB that can power the essential avionics and various projects on a fixed-wing aircraft.

The board is designed to be able to take in both a 12S and 4S input to eliminate the need for a dedicated avionics battery. There are various subcircuits on each board which are detailed in this document. This document serves to explain the overall design process of the PDB, covering requirements, schematic design, layout design and any iterations on circuits.

4 Typical Application



5 Technical Specifications

	Unit	Value
Input voltage	V	16.8–50.4
Maximum continuous current	A	0.05–10.56
Maximum power usage	W	112.66W
Output voltages	V	12V, 8V, 7V, 5V and 3.3V

Regulator	Name	IC	Max Current
A	12V OUT 1	LM70880-Q1	8A
B	12V OUT 2	LM70880-Q1	8A
C	5V RDNT	LMR36520FADDAR	2.5A
D	8V (Servo Right)	TPS56837	8A
E	8V (Servo Left)	TPS56837	8A
F	7V	LMR51450	5A
G	5V	LMR51450	5A
H	3.3V	TPS7A20	0.5A

6 Pinout

Pin	Signal	Connector Type	Recommended Max Current Draw
1	VBUS	Bus Bar (Input)	N/A
2	VIN	XT60 (Input)	N/A
3	5V RDNT	6 Pin Clickmate 2.00 MM	1.510A
4	12V	6 Pin Clickmate 2.00 MM	12.750A
5	8V (Servo Right)	6 Pin Clickmate 2.00 MM	6.250A
6	8V (Servo Left)	6 Pin Clickmate 2.00 MM	6.250A
7	7V	6 Pin Clickmate 2.00 MM	0.75A
8	5V (Cube)	6 Pin Clickmate 2.00 MM	1.215A
9	5V (Raspberry Pi)	2 Pin Clickmate 2.00 MM	2.5A
10	3.3V	2 Pin Clickmate 2.00 MM	0.125A

7 Measures

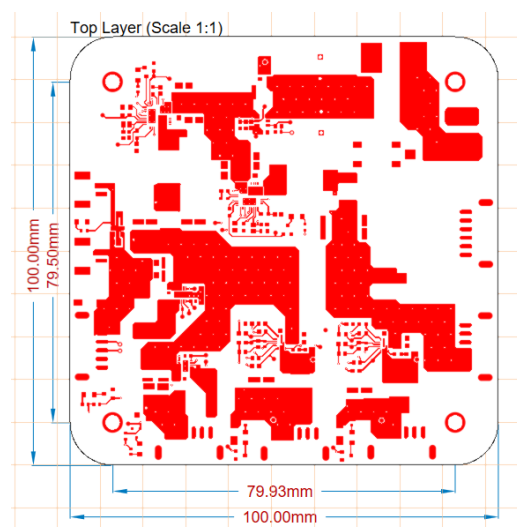


Figure 1: Measured size of PDB

8 Schematics

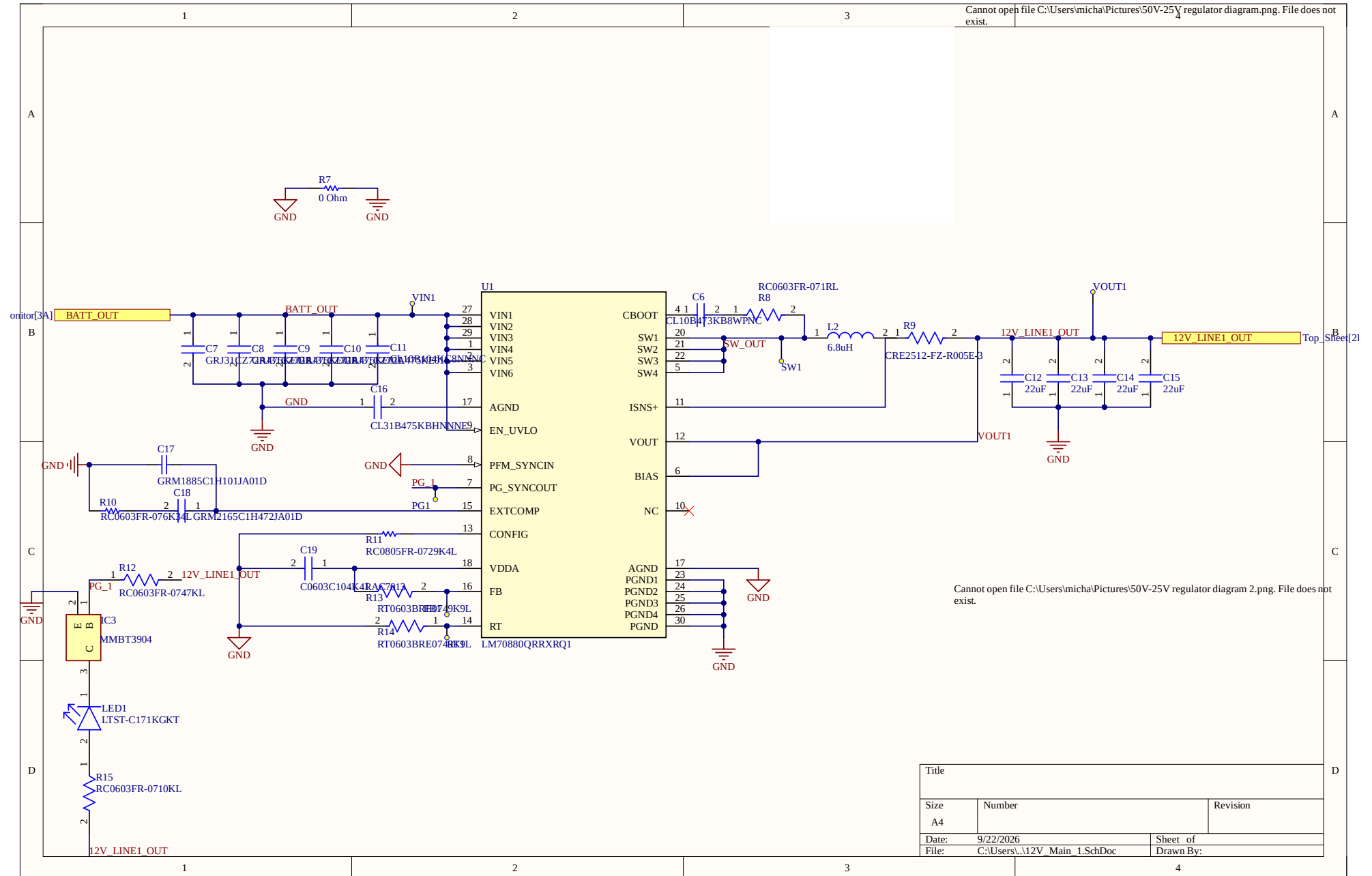


Figure 2: 12V main regulator, line 1 (LM70880-Q1)

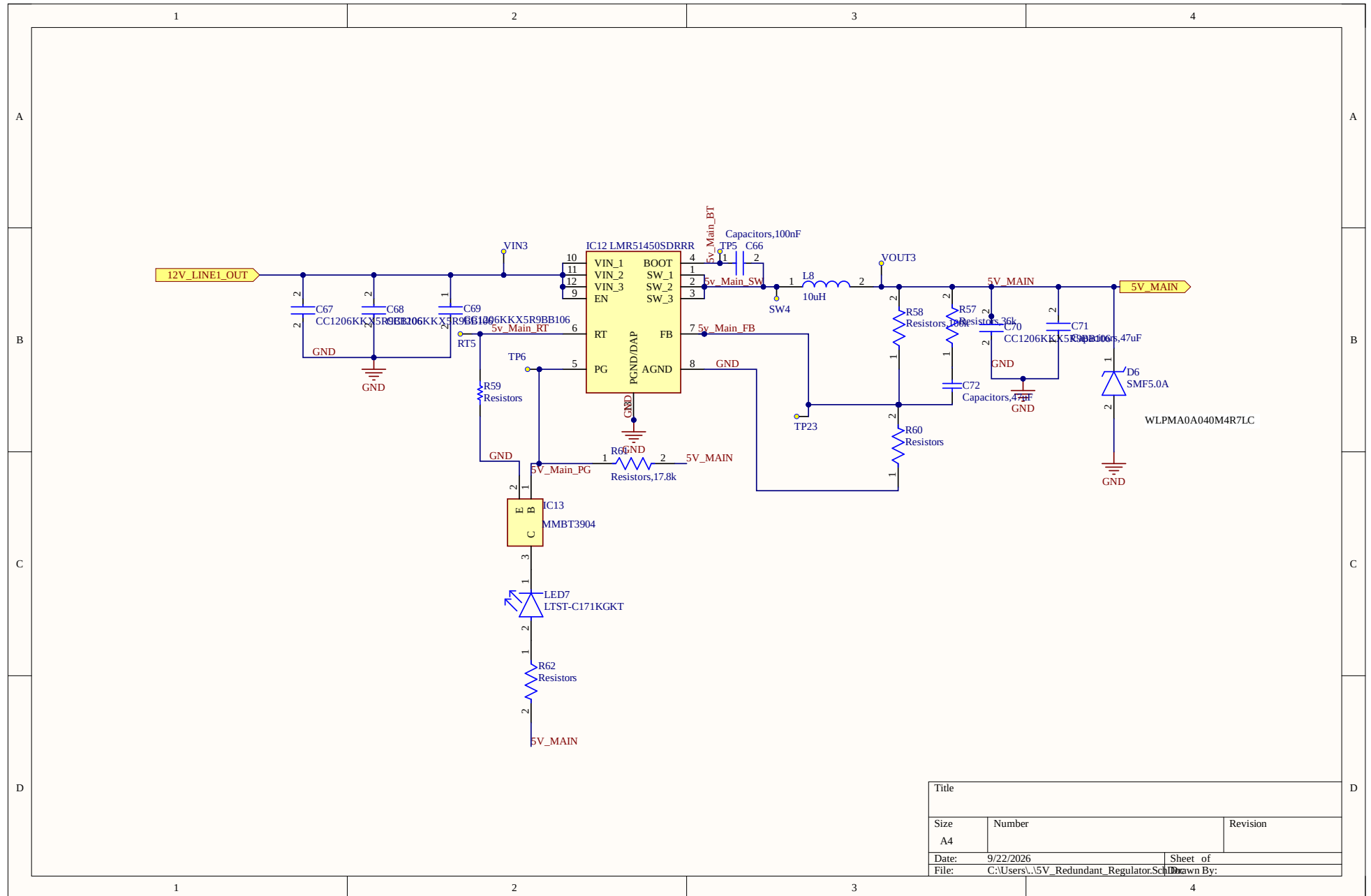


Figure 4: 5V redundant regulator (LMR51450)

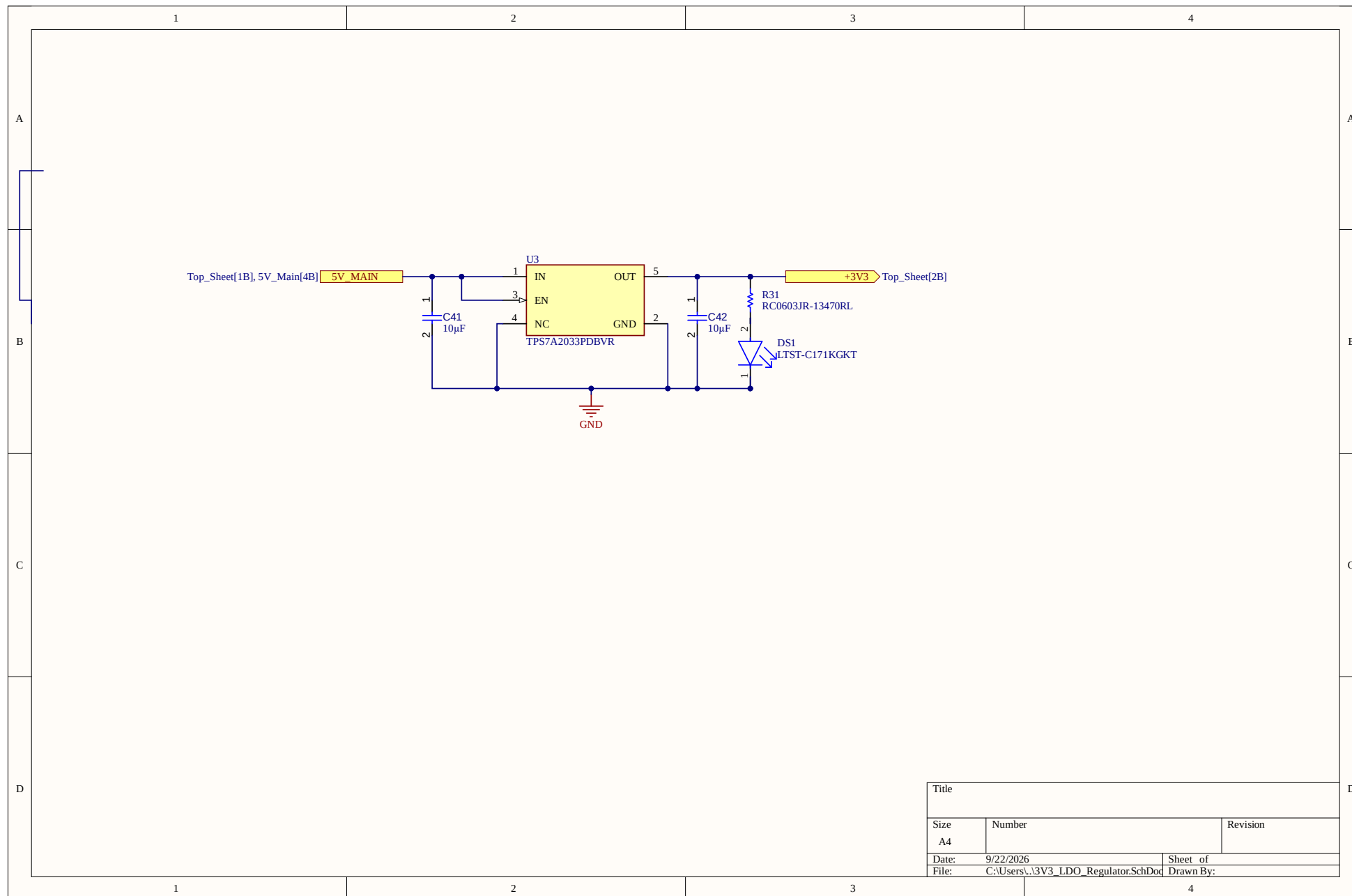
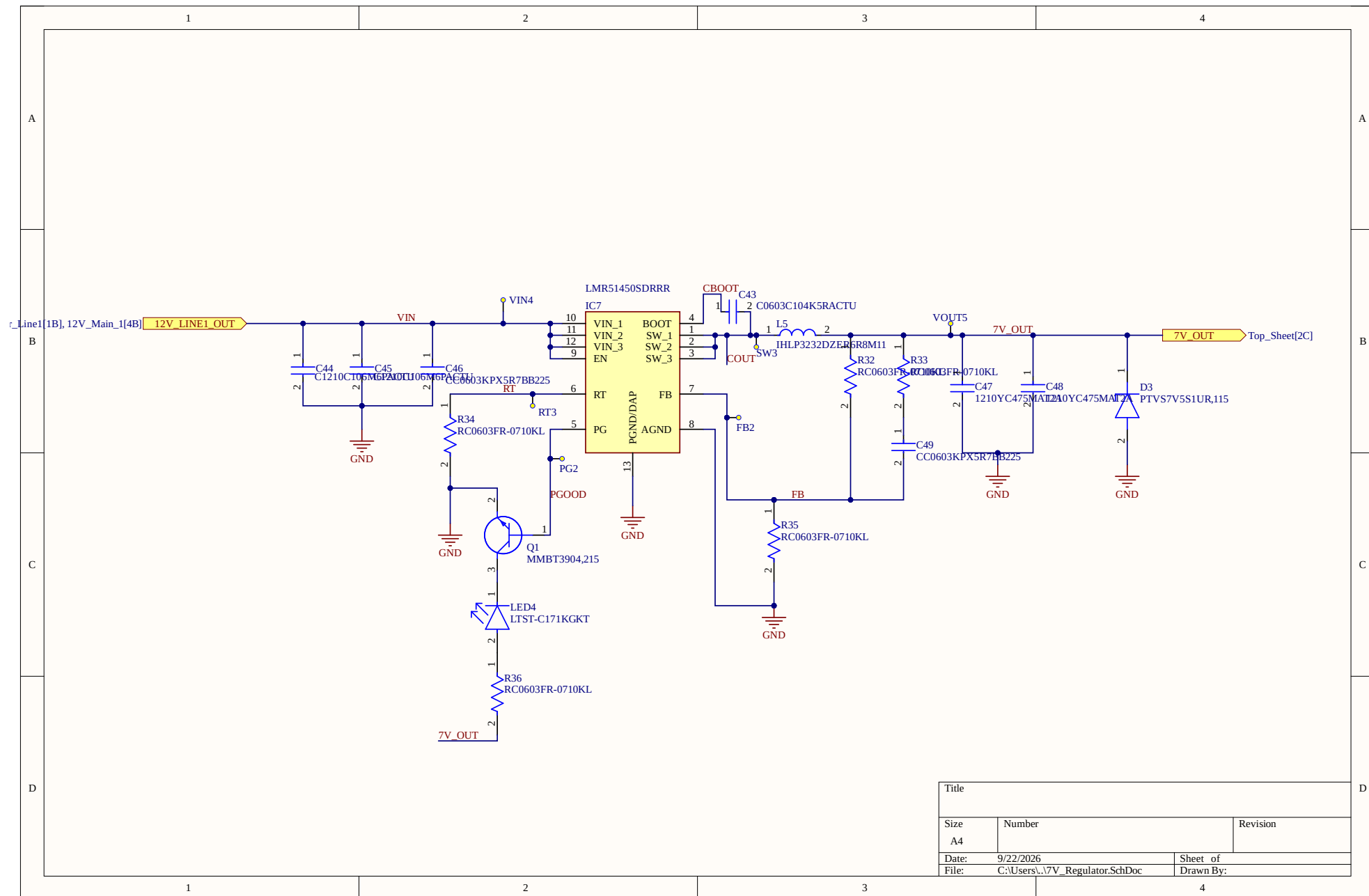


Figure 5: 3.3V LDO regulator (TPS7A20)



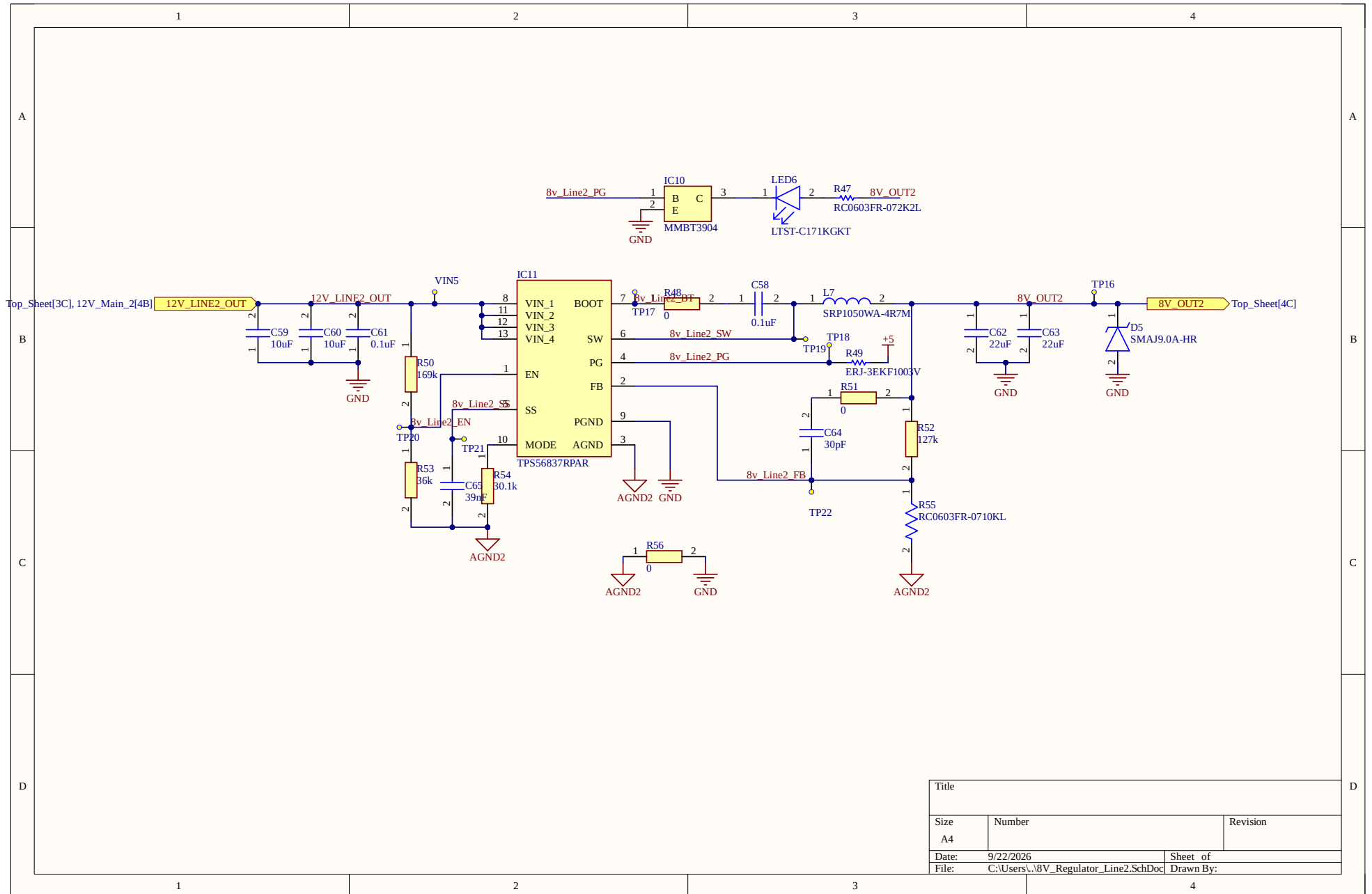


Figure 8: 8V regulator, line 2 (TPS56837)

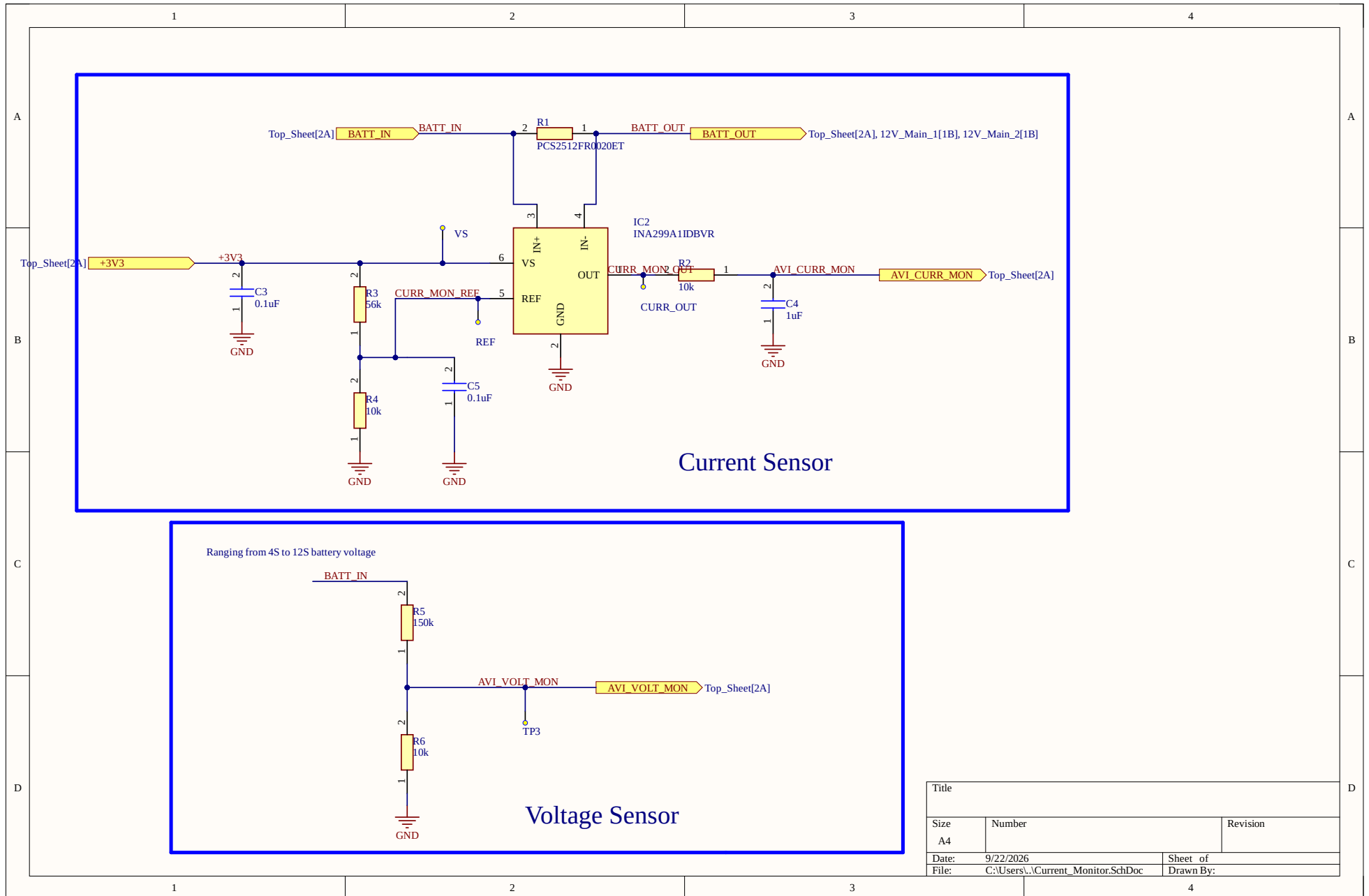
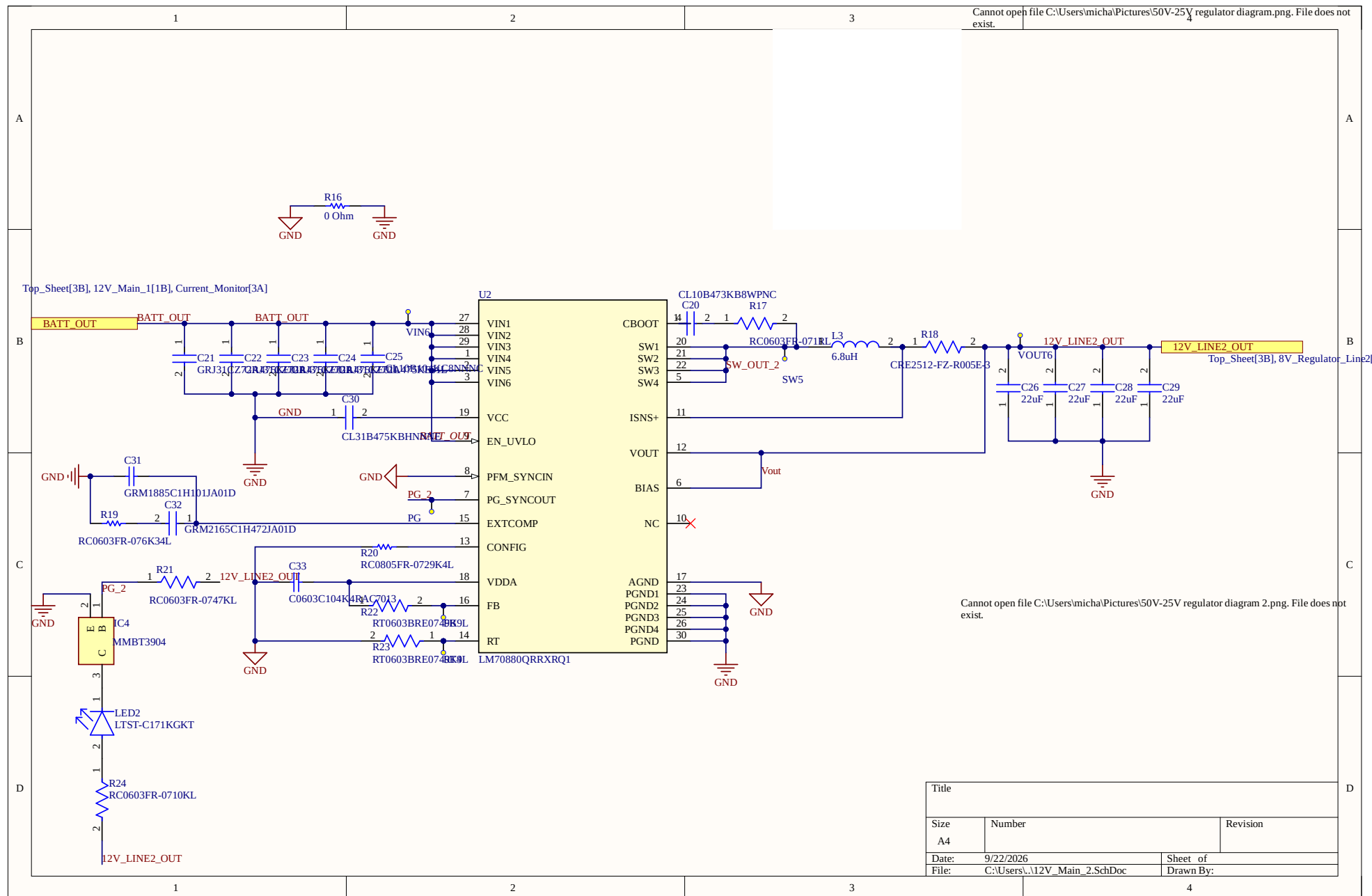


Figure 9: Current and voltage monitor



9 Acknowledgements

Designed and manufactured By: Michael Wang, Izaak Estandarte, Justin Wu, Dylan Ngere